

THE LIVES OF STARS

ALL STARS START LIFE AS HOT BALLS OF GAS THAT HAVE CONTRACTED DOWN FROM LARGER CLOUDS OF GAS AND DUST UNDER THE INFLUENCE OF GRAVITY. WHAT HAPPENS TO A STAR NEXT DEPENDS ON ITS INITIAL MASS.

Stars that form from the smallest clumps of gas and dust become relatively small, cool objects known as red dwarfs. These are the most common stars in our galaxy and last for tens of billions to trillions of years. As red dwarfs age, it is theorized that their surface temperature and brightness increase until eventually they become objects called blue dwarfs, then white dwarfs. Finally they fade to cold, dead, black dwarfs. However, the Universe is not yet old enough for even a blue dwarf to have formed.

Lives of medium- and high-mass stars

Medium-mass stars (about the size of the Sun) have shorter lives than red dwarfs, lasting for billions to tens of billions of years. They swell into red giants at the end of their lives. A red giant eventually sheds its outer layers to form an object called a planetary nebula, together with a hot, compact star remnant, known as a white dwarf. The very largest stars have the shortest lives, measured in millions to hundreds of millions of years, because they use up their hydrogen fuel very quickly. In time, they form red supergiants, which disintegrate in stupendous explosions called supernovas. Depending on its mass, the core left by a supernova shrinks to one of two bizarre objects: a neutron star (see pp.36–37) or a stellar black hole (see pp.38–39).

